TECHNICAL DATA SHEET & PACKAGE INSERT

Dataset: NRSur7dq4v2 Precessing Binary Black Hole Waveforms & Frame Dynamics

Version: 1.2 (2026)
Total Size: 12.59 GB (11.72 GiB)
Format: HDF5 / Python 3.11

Executive Formulation: High-fidelity precessing gravitational waveform ensemble ($N = 102\,400$ samples, $L = 2048$ time steps) generated via the numerical relativity surrogate model `NRSur7dq4v2`. Intended for training autoencoders, surrogate emulators, and parameter estimation networks. The entire dataset is strictly normalized to dimensionless geometric units ($G = c = M_{\rm tot} = 1$), where time t is expressed in units of total mass M, and amplitude is given as dimensionless strain rh_{22}/M.

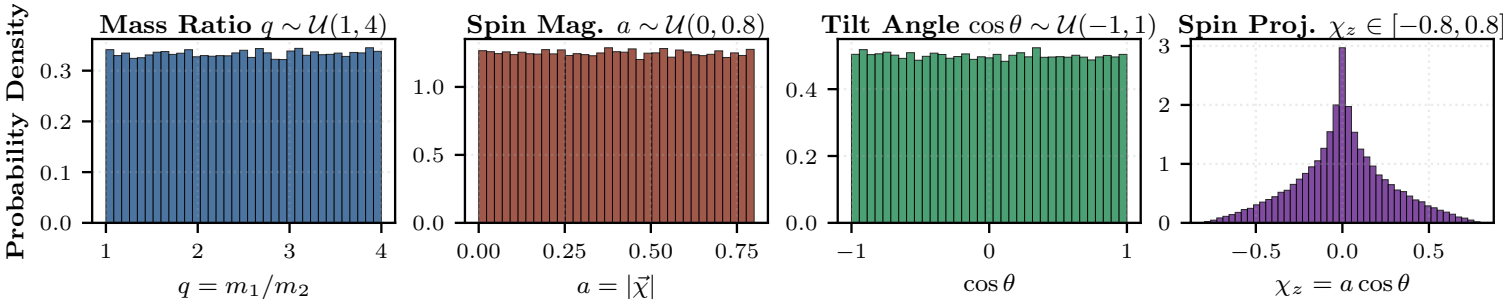


Fig 1: Parameter distributions across 100k samples: flat mass ratio q, flat spin magnitude a, isotropic orientation $\cos\theta$, and centered projection χ_z.

1. DESIGN SCOPE & STORAGE RATIONALE

Storage Footprint Justifications (Single Mode & Inertial Only):

- **Single Mode Only ($(2,2)$):** A complete precessing system radiates into all $\ell = 2$ modes ($m \in \{\pm 2, \pm 1, 0\}$). Storing all 5 modes would increase waveform size from 3.36 GB to 16.8 GB (total dataset $>$ 26 GB). The $(2,2)$ mode contains $>$ 85–90% of total emitted power and preserves the dominant precession modulation.
- **Inertial Frame Exclusively (No Coorbital Duplication):** Coorbital modes are non-precessing and smooth. Storing coorbital waveforms alongside inertial ones would double waveform storage (+3.36 GB). Instead, the full continuous transformation is preserved compactly via the unit quaternion $\mathbf{q}_{\rm copr}(t)$ and $\phi_{\rm orb}(t)$ in `/Dynamics`.

2. HDF5 ARCHITECTURE & DATA TYPES

HDF5 Dataset Path	Tensor Shape	Type	Size
`/Waveforms`	$(102400, 2048)$	`c128`	3.36 GB
`/Parameters`	$(102400, 7)$	`f32`	2.87 MB
`/Dynamics/q_copr`	$(102400, 4, 2048)$	`f32`	3.36 GB
`/Dynamics/chiA`	$(102400, 2048, 3)$	`f32`	2.52 GB
`/Dynamics/chiB`	$(102400, 2048, 3)$	`f32`	2.52 GB
`/Dynamics/orbphase`	$(102400, 2048)$	`f32`	0.84 GB
Total Structure	6 Datasets (1 Root, 1 Group)		**12.59 GB**

Temporal Grid Properties:

- Grid: $t \in [-3996.0, +98.0]\,M$, uniform step $\Delta t = 2.0\,M$ ($L = 2048$).
- Peak Alignment: Merger epoch $t = 0.0\,M$ is exactly at index **1998** ($\sim 97.6\%$), leaving 50 post-merger samples ($100\,M$) for ringdown.

3. PARAMETER PRIORS & SAMPLING LAWS

- **Mass Ratio:** Uniform $q = m_1/m_2 \sim \mathcal{U}(1.0, 4.0)$ ($m_1 \geq m_2$).
- **Spin Magnitudes:** Uniform length $a_1, a_2 \sim \mathcal{U}(0.0, 0.8)$.
- **Directional Isotropy ($p(\Omega) = 1/4\pi$):** Spin orientations are uniformly distributed across the full sphere surface (total solid angle $\Omega = 4\pi$ sr), meaning zero directional bias in 3D:

$$\cos\theta \sim \mathcal{U}(-1.0, 1.0), \quad \phi \sim \mathcal{U}(0.0, 2\pi), \quad p(\Omega) = \frac{1}{4\pi}\ \text{sr}^{-1}$$

- **Cartesian Projections:** Evaluated into 3D vectors:

$$\chi_x = a\sin\theta\cos\phi, \quad \chi_y = a\sin\theta\sin\phi, \quad \chi_z = a\cos\theta$$

4. GLOBAL DATASET STATISTICS

Empirical statistics computed across $N = 102\,400$ for neural network preprocessing (e.g. PyTorch / Autoencoders in `train2AE.py`):

Variable	Min	Max	Mean μ	Std σ
Mass Ratio q	1.000	4.000	2.500	0.866
Spin Components $\chi_{A,B}^{\{x,y,z\}}$	-0.800	+0.800	0.000	0.270
Waveform Amplitude $\lvert h_{22}\rvert$	0.000	0.385	0.028	0.046
Real / Imag $\mathrm{Re}(h), \mathrm{Im}(h)$	-0.385	+0.385	0.000	0.038
Waveform Phase $\phi = \arg(h)$	$-\pi$	$+\pi$	0.000	1.814
Quaternion $\mathbf{q}_{\rm copr}$ (components)	-1.000	+1.000	—	—
Orbital Phase $\phi_{\rm orb}$ [rad]	0.000	152.4	45.2	32.1

5. EVALUATOR SETTINGS & EVALUATION CALL

Waveforms are generated using `NRSur7dq4v2` via the specific call:

```
sur(q=q, chiA0=chiA0, chiB0=chiB0, f_low=0.0,
times=times, ellMax=2, precessing_opts=precessing_opts)
```

Detailed Parameter Specifications:

- `times` **(Temporal Evaluation Grid):** A 1D array of 2048 uniform time values:

```
times = np.arange(-3996, 100, 2.0)
```

  Spans from $-3996\,M$ (early inspiral) to $+98\,M$ (post-merger ringdown). Because $t = 0.0\,M$ is the merger peak, index **1998** aligns exactly with the peak, allocating 50 samples for quasinormal ringdown.
- `precessing_opts` **(Precession Control Dictionary):**
  - `'init_orbphase'`: `0` → Fixes initial orbital phase to 0 at reference time t_0, placing the separation vector $\hat{n} = \text{body}_2 \to \text{body}_1$ along the coordinate $\hat{x}$-axis.
  - `'init_quat'`: `[1, 0, 0, 0]` → Unit quaternion identity. Fixes the reference gauge so that at $t = t_0$, the coprecessing frame axes coincide directly with the inertial frame ($\hat{z}_0 \parallel \vec{L}_0$).
  - `'return_dynamics'`: `True` → Instructs the surrogate to output the underlying dynamical evolution dictionary: quaternion $\mathbf{q}_{\rm copr}(t)$, spin vectors $\vec{\chi}_A(t), \vec{\chi}_B(t)$, and accumulated orbital phase $\phi_{\rm orb}(t)$.
- `f_low=0.0` **&** `ellMax=2`**:** Starts from initial orbital frequency without artificial clipping, extracting quadrupole modes up to $\ell = 2$.

6. FRAME DYNAMICS & MATHEMATICAL TRANSFORMATIONS

Precessing binary dynamics naturally evolve through three distinct spatial coordinate representations:

1. **Coorbital Frame (Orbit-Following):**
   - $\hat{x}(t)$ points from body 2 to body 1 along $\hat{n}(t)$; $\hat{z}(t)$ aligns with instantaneous orbital angular momentum $\hat{L}(t)$.
   - *Waveform morphology:* Waveforms in this frame are non-precessing, smooth chirps where over 99% of the energy is concentrated in the $(2, \pm 2)$ mode without precession dips.
2. **Coprecessing Frame (Precession-Tracking / Minimal Rotation):**
   - Tracks the precession of $\hat{L}(t)$ across space, but does not rotate at the rapid orbital frequency $\omega_{\rm orb}$.
   - *Transformation from Coorbital:* Given the orbital phase $\phi_{\rm orb}(t)$ stored in `/Dynamics/orbphase`, modes rotate around $\hat{z}$ as:

$$h^{\rm copr}_{\ell,m'}(t) = e^{-im'\phi_{\rm orb}(t)}\, h^{\rm coorb}_{\ell,m'}(t)$$

3. **Inertial Frame (Laboratory / Radiation Frame):**
   - Fixed in space. The orbital plane wobbles relative to this frame, producing characteristic amplitude modulation and frequency shifts.
   - *Transformation from Coprecessing:* The time-dependent unit quaternion $\mathbf{q}_{\rm copr}(t) = [q_0, q_1, q_2, q_3]$ stored in `/Dynamics/q_copr` parameterizes the spatial 3D rotation via Wigner $\mathcal{D}$-matrices:

$$h^{\rm inertial}_{\ell,m}(t) = \sum_{m'=-\ell}^{\ell} \mathcal{D}^{\ell}_{m,m'}(\mathbf{q}_{\rm copr}(t))^*\, h^{\rm copr}_{\ell,m'}(t)$$

Inversion Principle for ML: An autoencoder trained on smooth coorbital modes can reconstruct full inertial waveforms by applying $\phi_{\rm orb}(t)$ and $\mathbf{q}_{\rm copr}(t)$ using the forward Wigner $\mathcal{D}$-matrix equation above.